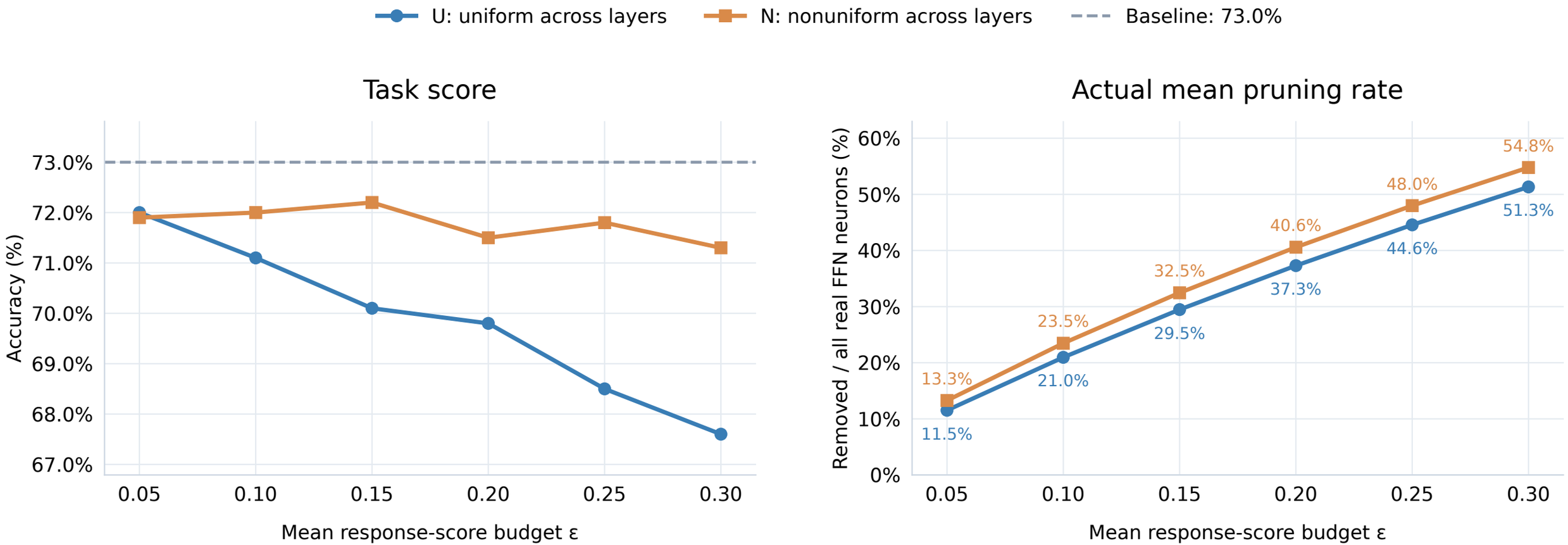
Uniform vs nonuniform pruning at the same response budget

Gemma-4-E2B-it | MMBench | Sample-adaptive selection

Measured results | N = 1000 paired samples per condition



Actual mean response-score share removed (B)

Method	ε=0.05	ε=0.10	ε=0.15	ε=0.20	ε=0.25	ε=0.30
U: uniform	4.9971%	9.9969%	14.9965%	19.9963%	24.9960%	29.9958%
N: nonuniform	4.9999%	9.9999%	14.9999%	19.9999%	24.9999%	29.9999%

Budget = equal-layer mean response share. Pruning rate = total removed / total neurons.

Task-score axis is zoomed. More removal does not guarantee better accuracy.